



D1 DNA replication

D · Continuity and change · IB Diploma · Concept guide and original exam-style practice

Understand the concept

DNA replication is semi-conservative: the enzyme helicase unwinds and separates the two strands of the double helix by breaking hydrogen bonds between base pairs, and DNA polymerase then synthesises a new complementary strand alongside each original (template) strand, adding nucleotides in the 5'→3' direction according to complementary base pairing. Because DNA polymerase can only extend an existing 3'-OH end, a short primer is needed to begin synthesis; the leading strand is synthesised continuously toward the replication fork, while the lagging strand is synthesised discontinuously, away from the fork, as a series of Okazaki fragments later joined by DNA ligase. The result is two DNA molecules, each containing one original strand and one newly made strand.

Worked example: follow the reasoning

The Meselson-Stahl-style logic testing semi-conservative replication: bacteria are first grown in medium containing only heavy nitrogen (15N), so all their DNA is 'heavy'. They are then transferred to medium with only light nitrogen (14N) and allowed to replicate once. Step 1: under the semi-conservative model, each new DNA molecule should contain one original heavy (15N) strand and one newly made light (14N) strand, giving DNA of intermediate ('hybrid') density. Step 2: under a (disproved) conservative model, replication would instead produce one fully heavy and one fully light molecule, with no intermediate-density DNA. Step 3: when DNA was extracted after one round of replication and its density measured (e.g. by density-gradient centrifugation), only a single intermediate-density band was found, matching the semi-conservative prediction. Conclusion: this result supports the semi-conservative model of DNA replication.

Quick check · multiple choice

Which enzyme is primarily responsible for unwinding the DNA double helix at the replication fork?

- A DNA polymerase
- B Helicase
- C DNA ligase
- D Primase

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 During DNA replication, the two new DNA strands are synthesised differently: one continuously and one in short discontinuous fragments. [6]
 - (a) Explain why DNA replication is described as 'semi-conservative'. [2]
 - (b) Explain why the leading strand can be synthesised continuously, but the lagging strand must be synthesised discontinuously as a series of Okazaki fragments. [3]
 - (c) State the role of DNA ligase in this process. [1]

Final answer: _____



Answers and explanation

Question 1 [6]

(a) Each of the two new DNA molecules produced consists of one original (parental) strand and one newly synthesised strand; because half of each original molecule is conserved in each daughter molecule, the process is called semi-conservative.

(b) DNA polymerase can only add nucleotides to the 3' end of an existing strand, synthesising in the 5'→3' direction; on the leading strand, this direction of synthesis matches the direction the replication fork opens, so synthesis is continuous. On the lagging strand, the template runs in the opposite orientation relative to fork movement, so DNA polymerase must synthesise short segments back toward the fork each time more template is exposed, producing a discontinuous series of Okazaki fragments, each requiring its own primer.

(c) DNA ligase joins the Okazaki fragments together (by sealing the sugar-phosphate backbone between adjacent fragments) to form a continuous lagging strand.

Quick check answer

B. Helicase breaks the hydrogen bonds between complementary base pairs, unwinding and separating the two DNA strands so each can act as a template.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

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